

Multiple Choice

1. **Answer:** D

Options: A) Iron; B) Hydrogen; C) Methane; D) Helium

Note: Correct option: D - Helium

2. **Answer:** B

Options: A) Being hit in the head by a bullet.; B) Being hit by a small meteorite.; C) Starvation during global winter caused by a major impact.; D) Driving while intoxicated without wearing seatbelts.

Note: Correct option: B - Being hit by a small meteorite.

3. **Answer:** D

Options: A) From the layering of materials within the Earth.; B) From fossils of ancient life.; C) From the cratering history of Earth's surface.; D) From radioactive dating of rocks and meteorites.

Note: Correct option: D - From radioactive dating of rocks and meteorites.

4. **Answer:** D

Options: A) the softer rocky material of the mantle.; B) the lava that comes out of volcanoes.; C) material between the crust and the mantle.; D) the rigid rocky material of the crust and uppermost portion of the mantle.

Note: Correct option: D - the rigid rocky material of the crust and uppermost portion of the mantle.

5. **Answer:** D

Options: A) patterns of motion; B) the presence of asteroids and comets; C) the measured ages of meteorites; D) the presence of life on Earth

Note: Correct option: D - the presence of life on Earth

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: '150 million km'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Its rotation axis is nearly perpendicular to the plane of the Solar System.'

8. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: '7'.
9. **Answer:** True
Options: A) True; B) False
Note: LLM-generated true statement based on MCQ.
10. **Answer:** False
Options: A) True; B) False
Note: LLM-generated false statement. Correct answer: 'About 6 meters'.

Long Form Response

11. **Answer:** Jupiter's greater mass compresses it more thus increasing its density.. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.
12. **Answer:** Mass 5 kg weight 25 N. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.
Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.